PH-UY 2023: ELECTRICITY, MAGNETISM, & FLUIDS

FINAL EXAM
5/12/25

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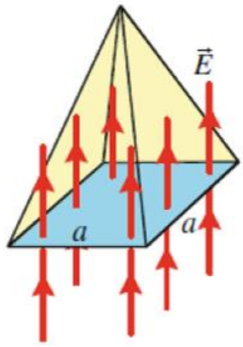
BEFORE YOU START…

On the exam, blue book, and answer sheet, write:
- your FULL name (as listed by the university registrar)
- your NetID (i.e. your NYU email minus "@nyu.edu")

1. A beaver floats on a log as it rides down a river. If the beaver has a mass of 20 kg, and 15% of the log's volume is *above* the water, what is the volume of the log? Assume the specific gravity of the log is 0.7. Recall that specific gravity is the ratio of the density of a material to the density of water.

a) 0.133 m^3 b) 0.167 m^3 c) 0.111 m^3 d) 0.222 m^3 e) 0.194 m^3

2. A pyramid with a horizontal square base with sides of length *a* = 5 m is placed in a uniform electric field of 5200 N/C directed vertically upwards. There is no charge inside the pyramid. Find the total electric flux through any *one* of the pyramid's 4 identical slanted surfaces.

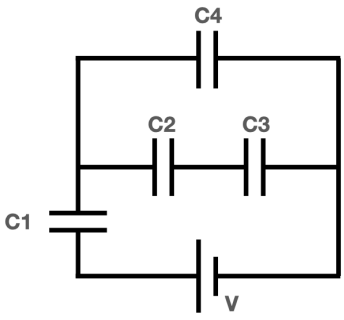


a) 1.03 x 10^4 N·m^2/C
b) 6.37 x 10^4 N·m^2/C
c) 4.68 x 10^4 N·m^2/C
d) 8.32 x 10^4 N·m^2/C
e) 3.25 x 10^4 N·m^2/C

3. Three point charges, each with charge +q, are located at the corners of an equilateral triangle with sides of length 8L. The work required to move one of the charges from its corner to the middle of the opposite side and place it there at rest is Bkq^2/L, where B is equal to:

a) 1 b) 1/4 c) 1/2 d) 1/3 e) 1/6

4. In the circuit shown, V = 6 V, and $C_1 = C_2 = C_3 = C_4 = 6\ \mu F$. What is the charge on capacitor C_1?



a) 32.4 **μ**C
b) 54.0 **μ**C
c) 43.2 **μ**C
d) 21.6 **μ**C
e) 27.3 **μ**C

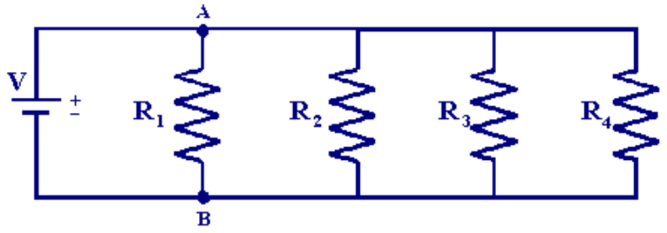
5. A 12 nF air-filled parallel plate capacitor is connected across a 75 V power supply until the capacitor is fully charged. The power supply is then removed, and a dielectric with constant 3.5 is inserted between the plates. How much energy stored by the capacitor is lost when the dielectric is inserted?

a) 18.1 μJ b) 11.2 μJ c) 8.68 μJ d) 13.0 μJ e) 24.1 μJ

6. A wire 10 m long with a cross-sectional radius of 0.6 mm has a resistance of 5 Ω. A second wire 3 m long with cross-sectional radius of 0.4 mm is made of the same material as the first wire. What is the resistance of the second wire?

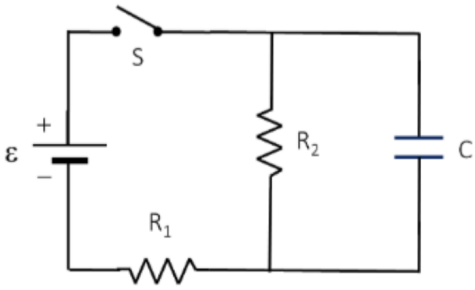
a) 6.00 Ω b) 9.10 Ω c) 3.38 Ω d) 2.16 Ω e) 13.5 Ω

7. In the circuit diagram, the battery has an emf of 18 V, and each resistor has an identical 12 Ω resistance. What is the total amount of heat generated solely by resistor R_1 in 1 minute?



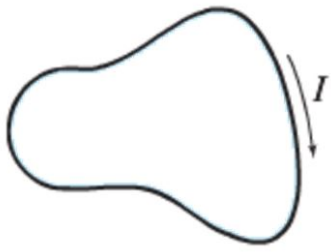
a) 1125 J b) 1620 J c) 2205 J d) 2880 J e) 3645 J

8. In the circuit shown, the C = 4 μF capacitor is initially uncharged, and the switch S starts open. The emf of the battery is 100 V, and the resistor values are R_1 = 5 Ω and R_2 = 15 Ω. A very long time after the switch is closed, what is the magnitude of the charge on each plate of the capacitor? (1 μF = 10^{-6} F)



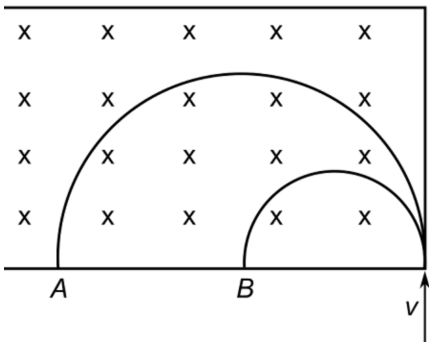
a) 220 μC b) 240 μC c) 260 μC
d) 280 μC e) 300 μC

9. The irregularly shaped loop in the figure lies entirely in the plane of the page and carries a steady clockwise current of 4 A. There is a uniform magnetic field of magnitude 5 T directed to the right of the page. If the loop experiences a torque about its center of 140 N·m at the instant shown, what is the area of the loop?



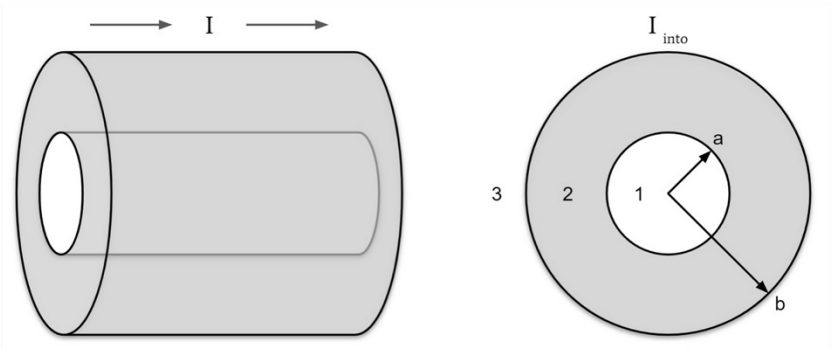
a) 7 m^2 b) 8 m^2 c) 9 m^2 d) 10 m^2 e) 11 m^2

10. The region shown (not drawn to scale) has a uniform magnetic field directed into the page. At $t = 0$, two particles moving perpendicular to the field enter the region at the lower right corner with the same unknown speed. **Particle A** has mass $9m$ and charge $+3q$, while **Particle B** has mass m and charge $+2q$. If Particle B reaches point B at $t = 0.5$ sec, at what value of t does Particle A reach point A?



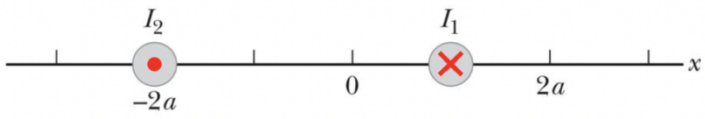
a) 2 sec b) 3 sec c) 4 sec
d) 5 sec e) 6 sec

11. An infinitely long straight hollow cylinder of inner radius a and outer radius $b = 3a$ carries a current I along its length, as shown (in the "End view", regions 1 and 3 are empty, and the currently is evenly distributed over the cross-sectional area in region 2). At a point a distance $(2/3)a$ from the central axis of the cylinder, the magnitude of the magnetic field is $\mu_0 I / (2 \pi a)$ times



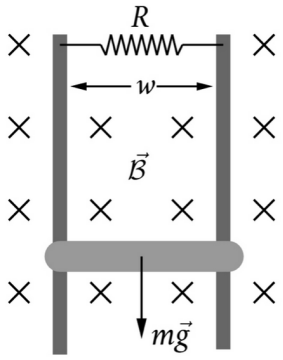
a) 0 b) 4/9 c) 9/16 d) 16/25 e) 49/64

12. An infinitely long straight wire passing through $x = a$ carries a current I_1 directed into the page. Another identical wire passing through $x = -2a$ carries a different current I_2 directed out of the page. If the magnitude of the magnetic field at $x = 0$ is $9 \mu_0 I_1 / (2 \pi a)$, what is the ratio I_2 / I_1?



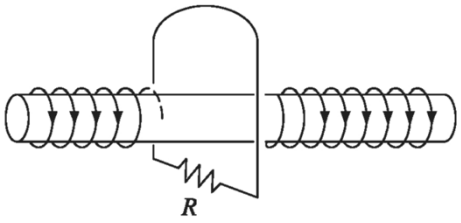
a) 1/1 b) 8/1 c) 12/1 d) 16/1 e) 20/1

13. A vertically conducting frame connected to an R = 72 Ω resistor is in a uniform magnetic field of magnitude B = 2 T directed into the page. A horizontal wire of length w = 3 m and mass m = 1.3 kg can slide without friction on the frame's rails, as shown in the figure. The wire is observed to fall at a constant speed and never loses contact with the rails. What is that speed? *(Use g = 10 m/s^2 to simply calculation.)*



a) 26 m/s b) 28 m/s c) 30 m/s d) 32 m/s e) 34 m/s

14. A long solenoid with radius (2/π) m and 15 turns per meter is surrounded by a wire loop with area 3 m^2 and resistance R = 12 mΩ, as seen in the figure. If the current in the solenoid is increasing at a steady rate of 1.6 A/s, what is the induced current in the wire loop?



a) 2.4 mA b) 2.8 mA c) 3.2 mA d) 3.6 mA e) 4.0 mA

15. A battery is connected to a 250 mH inductor, a 5 kΩ resistor, and a switch, all in series. The switch is initially open. How long after the switch is closed will the current in the inductor be 55% of the value it reaches after the switch has been closed for a long time?

a) 69.3 μs b) 52.5 μs c) 72.1 μs d) 39.9 μs e) 94.9 μs